

Institute Of Actuaries Of India

Subject CA1 –Paper II

May 2011 Examinations

INDICATIVE SOLUTION

Q.1)

Reinsurance can be used to reduce the uncertainty associated with timing and amount of mortality related claims. In particular:

- i. Reinsurance can be used to reduce claims volatility and hence smooth profits and reduce capital requirements
- ii. Reinsurance can be used to limit exposure to risk. For eg., limiting losses from a single risk or a single claim, cumulative events and impact of concentration of risk
- iii. Reinsurer could provide expertise in pricing- in setting mortality assumptions for products where company has own limited experience
- iv. Assist in wording of policy documents- eg exclusion clauses
- v. It can assist in setting up underwriting, and claims process and provide periodic review of the same
- vi. Help the insurer diversify; as the insurer retains smaller proportion of large number of risks. Hence the insurer can write a wider range of risks
- vii. Offer assistance to insurer in underwriting and accepting very large risks

[Total Marks – 5]

Q.2)

The key factors to be considered are:

- i. Asset share
 - Asset share is the retrospective accumulation of the premiums less expenses and cost of cover at the actual rate of return earned on the assets
 - For any contract this should be starting point for setting discontinuance terms. It will be the maximum the company can afford to pay
 - The company could actually pay more or less than this value, which may depend on durations
 - Another factor to consider is how does surrender value compare with reserving basis i.e. may want surrenders to be source of surplus or at least not a deficit
- ii. Policyholder expectations
 - At shorter durations, it is possible for the policyholder to compare it with the premiums paid (with some interest)
 - However the asset share at this stage is usually less than the premiums paid or sometimes even negative
 - The insurer might be willing to take some loss or reduced profit so that the surrender value does not appear too low compared to the premiums paid
 - Regulations may specify what to pay in these circumstances

- Closer to maturity- discontinuance value would be expected to be closer to the maturity value as per the contract.
- Discontinuance terms have to allow for PRE created by statements in policy/ sales literature etc.
- In respect of with profit business some allowance may be needed for any bonuses ie may want to pay more than just guaranteed benefits

iii. Competition

- What the competitors offer is equally important.

iv. Other practical considerations

Ease of calculation

- Cost of implementing systems to calculate discontinuance terms
- Frequency of change of the discontinuance terms.

[Total Marks - 6]

Q.3)

Part (a)

- i. Reserves are set up to provide funds to meet future liabilities i.e. to pay expected claims under the particular policies
- ii. To determine the liabilities to be shown in the provider's published accounts and reports
- iii. If separate accounts and reports have to be prepared for the purpose of supervision of solvency, to determine the liabilities to be shown in those accounts
- iv. To determine the liabilities to be shown in internal management accounts and reports of the provider
- v. To calculate discontinuance/ surrender benefits
- vi. To determine the excess of assets over liabilities and whether any bonuses /profit share could be awarded
- vii. To value benefit improvements for a pension scheme and/or set future contributions to a benefit scheme (if managed by the insurer)
- viii. To determine appropriate valuations for merger and acquisition or sale of a block of business
- ix. To set investment strategy e.g. split between fixed and real liabilities

Part (b)

Global reserves would be for risks which are perceived to be at portfolio level of the insurer and not at a policy level.

- i. Due to mismatch between assets and liabilities
 - Risks relating to mismatch between assets and liability cash flows- which in case of change in investment conditions may result in liabilities increasing more than the assets, hence affecting the solvency position of the insurer
 - Overall strategic investment strategy would be at group level not at policy level. Hence the risk is at global level so the reserve set aside at global level
- ii. Global reserves are usually held for options and guarantees because they tend to affect a portfolio all in one direction or another(interest rate guarantees biting) and are hence determined at a portfolio level
- iii. Similarly another example would be catastrophic reserve, though each policy is small to create any risk to the insurer but one catastrophic (earthquake, tsunami) event can lead to concentration of risk, which needs to be covered at portfolio level
- iv. Global reserves could be held to cover tax liabilities or contingent benefits to employees e.g. pensions.
- v. Likewise some regulatory bases may specify reserves in global terms eg cash amounts or related to total fund sizes.
- vi. Additional global reserves could be held to demonstrate financial strength to consumers or investors.
- vii. Other financial and non-financial risks for which global reserves could be held:
 - Credit risk – default of any third party (e.g. reinsurer)
 - Operational risk – anti selection, regulatory fines or compensation for mis-selling
 - AIDS, or other potentially significant but uncertain risks, reserve
 - Interest equalization reserve/smoothing reserve

[Total Marks - 9]

Q.4)**Part (a)**

It is important because

- i. It will enable company to charge premiums that will cover the expected level of initial and on-going expenses associated with writing and subsequently servicing the contracts
- ii. Understand the level of cross subsidy between new business and renewals, between different lines of business and management of the same
- iii. It is always better to know the theoretically correct expense loadings even if company is not able to implement it commercially ie to understand the cost implications of not charging the correct loadings
- iv. understand the actual costs of writing and servicing contracts and how these may vary, for example by distribution channel(brokers, own sales force, internet)

Part (b)

The classifications could be:

i. Fixed Vs Variable expenses

- a. All expenses in the long run vary, so the classification of fixed and variable are in the short run
- b. Fixed expenses are those that don't vary by business volume e.g Property costs, salary costs
- c. Variable expenses on the other hand are those might be related to the amount of business by one or more different measure

ii. Direct Vs Indirect expenses

- a. Direct expenses are those that are related specifically to a particular class or classes of business.
- b. Example; underwriting expense, commission, etc
- c. Indirect costs relate to the support functions such as: general management, human resources; IT
- d. All variable expenses are direct but fixed expenses can be direct or indirect.

iii. Split by function/activity

- a. Acquisition expense- related to securing new business,
- b. maintaining expense- related to policy serving and maintenance
- c. investment related and
- d. claims or termination expenses

Part (c)

Inclusion in premium rates

The results of such an expense analysis could be loaded in premium rates in terms of:

- Premium income : commission
- No of policies : policy document printing, postal costs
- No of claims : claims form pricing and postal costs
- Sum assured/Claim amount : legal or underwriting costs

The mechanism could be:

- i. Split total direct expenses by product or by type of cover; so that the same can be appropriately charged in the premium rates
- ii. Expenses related to number of policies or claims would need to be divided by an average policy or claim size and loaded into premium rates
- iii. In respect of fixed expenses related to procurement of new business, assumption needs to be made about volume of new business. The fixed expense are then split at a per policy level
- iv. Indirect expense should be apportioned across all lines of business and allocated on a per policy basis
- v. The apportionment could be based on some form of charging out basis.
- vi. For example by using time sheets, floor space or by an arbitrary split say in proportion to the direct expenses split.

[Total Marks - 12]

Q.5)**(a)****Immediate Annuity**

- An annuity provides a guaranteed income no matter how long you live for ie it insures against you outliving one's own savings.
- but if you invest yourself then there is no insurance, The longevity risk is with you, you may outlast your investments
- For persons with poor health, the statement might be true; but insurers provide impaired life annuities (which are higher)
- Investment and reinvestment risk is passed to the insurance company with the purchase of an immediate annuity
- That is annuity terms are fixed at the outset i.e. they don't depend on future investment conditions.
- The size of the insurers' funds means that they can diversify across assets to be able to pay a steady income.
- Trying to do the individually would be difficult since fixed interest securities give coupons and redemption amounts – i.e. harder to plan.
- It is true that the insurer will have margins for its expenses and profits but
- Insurer brings in investment expertise ie the benefits outweigh the costs.
- While the individual may not be financial savvy to decide where to invest in and it costs time and resources
- Also due to the size of the investment (fund size) the insurer must be getting a better deal compared to individual in the market
- The transaction cost also per unit might be high for the individual. That is you will also incur expenses
- If individual invest in corporate bonds then the credit risk lies with them whereas the annuity rates are guaranteed and credit risk lies with insurance company
- In particular the insurer can invest in lots of bonds and diversify whereas an individual couldn't hence one default is more serious for an individual
- It is possible that due to competitive pressure or to gain market share insurer prices below cost, in which case it is advantageous for customers
- If there is sufficient free assets, the insurer might also consider investing in equities and providing overall better returns
- There might be tax or regulatory advantage of purchasing the annuity from the insurer than direct purchase of assets

(b)

- Both products are priced for the same profit margin and the 2 options are cost neutral in actuarial and probabilistic terms
- That is, in general, customers get what they pay for.
- In this case the slightly higher premiums paid with 2 single life products reflects the slightly higher value of benefits
- You and your spouse are relatively young and the term of the policies are relatively short.
- Hence there is only a low chance of 1 death and a very, very low chance of 2 deaths during the policy term.
- This means that it is unlikely that any extra benefit would be payable if you took out the 2 single life policies.
- In your case, for the same benefit, you would pay 200 for 2 single life policies and 190 for the joint life policy. The lower cost of 10 is effectively the value of the chance of 2 deaths. That is the difference in value of potential benefits under the 2 routes is 10.
- Some expenses are related to per policy irrespective of its size. So processing 2 policies would incur more expenses. So the difference of in premiums (between joint life and 2 single policies) isn't just about value of benefits but also due to savings in expenses. (in the above e.g. its say 8 due to benefits and 2 extra expenses).
- So in effect less premium goes towards benefits on the 2 policies route. That is 2 policies are less efficient than a joint life policy.

(c)

- It is not appropriate to compare two health plans only on premiums and decide which is the better of the two
- This is because there are a lot of different factors to consider under such policies and these factors will vary between insurers
- In effect you get what you pay for ie in general, higher premium implies greater potential benefits
- our company's product provides a comprehensive range of benefits to cover a wide range of circumstances while company XYZ's product is designed to look at serious events only i.e. they target different markets with different needs

- It is necessary to compare our policy's terms and conditions with that of Company XYZ. We believe that in the following areas, our product is superior;
 - i. Waiting periods and exclusions could be different
 - ii. There could be various sub-limits for various specific treatments
 - iii. Other services like cashless facilities, network of hospitals available
 - iv. Conditions relating to No claims discounts,
 - v. term of payment restrictions,
 - vi. indexation of benefits
 - vii. eligibility of benefits relating to pre-existing conditions plus
 - viii. efficiency and customer service type issues

give marks for any other valid issues here
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- Underwriting standards of the 2 companies are different as
- Some companies underwrite at the claims stage; so you can't be sure of the payment from the policy for treatment. No point paying lower premium when you are not sure if the claim would be paid when needed
- Hence overall, we believe that our comprehensive product does provide value for money. But, if it would be more suitable to your needs, we do offer cheaper but less comprehensive policies.

[Total Marks - 14]

Q.6)

Part (a)

Professional issues to be considered before accepting assignment-

- i. Whether you have the requisite qualification and experience to carry out the assignment
- ii. Check whether there is any conflict of interest, for eg.,
 - a. any close relative working for the company,
 - b. do you have significant shareholding in the company or
 - c. are you advisor to any investment fund which has significant holding in the company
 - d. are you or your firm advising the trustees or members of the scheme
- iii. Any conflict of interest should be disclosed up front or if you feel that the conflict is significant you should not accept the assignment.
- iv. In addition appropriate safeguards should be put in place to protect information (eg. Chinese walls).
- v. As it's a new advisory role for a one-off task, it should be ensured before accepting the assignment that sufficient information is available/can be provided for completing the task and timescale proposed is appropriate.

Part (b)

Suggestions that should be covered in the report

- i. Re look at Level and form of benefits
 - Consider making changes in the DB scheme in terms of accrual of benefits
 - Use average salary rather than the final salary for defining the pension
 - Exclude bonuses etc from pensionable pay or index pensionable pay to say inflation (ie less than actual pay).
 - Increase the normal retirement age; this may only defer the pension by a few years
 - Reduce the accrual rate of the pension;
 - However, benefit changes will only really work for future service. Trying to reduce past service benefits would reduce costs but it would be legally and morally difficult
- ii. Conversion to a DC scheme
 - Move from a final salary scheme to a defined contribution scheme will completely remove volatility in contribution rate
 - This switch from DB to DC all of a sudden might be too drastic a switch;
 - Hence a compromise is a hybrid scheme
 - But there are accrued benefit issues here as a transfer to a DC scheme won't be popular. So the employer could end up having 2 types of schemes –a DB for accrued benefits, a DC for new employees and (may be) future contributions of existing members diverted to a DC scheme. This could be messy and costly
- iii. Combine scheme benefits with state benefits; don't duplicate the benefits
- iv. Examine leaving service benefits , should be kept as low as legislation allows
- v. Keep transfer values cost neutral ie no enhancements eg on redundancy.
- vi. Consider insuring death in service benefits or even some active past service liabilities this will provide certainty in out go, at least
- vii. Currently scheme is well funded so could use surplus to reduce contributions
 - but surplus could be transitory which could cause problems later – reduce stability if contributions have to go up again later.
- viii. Also possible issues with legislation, scheme rules, members expectations and tax especially if surplus is repaid to employer or used to take contribution holiday/reduce future contributions.
- ix. Check what competition is doing
- x. Review benefits taken on ill-health or early retirement. If the benefits currently are generous, this could be curtailed
- xi. Review Options and guarantees
 - In case pension increases are promised; make them discretionary rather than guaranteed
 - Check out the options(for eg.,spouse pension) cost of all the options available under the scheme

- Check whether cost of options and guarantees are designed cost neutral if possible ie designed so that they aren't worth much
- xii. Consider making the eligibility conditions more stringent; have minimum number of years of service for joining the scheme, these might be part of the employment conditions
- xiii. Make it contributory scheme – let the employee contribute too; incase already there then increase employee contributions
- xiv. Look at the current asset management strategy
 - Do ALM study to ensure matching where required but taking calculated risks where appropriate.
 - Improve diversification to reduce volatility of returns
- xv. Funding of the scheme
 - Review the funding method used under the scheme, whether it provides stability
- xvi. Purchase deferred and immediate annuities from an insurance company to transfer longevity risk
 - This will reduce the risk but might not be a cheaper option
 - Get competitive quotes from various insurer to get the best deal
- xvii. Manage scheme expenses
 - Check whether you are getting value for money or
 - Look at going for 3rd party administration instead of employing staff to manage the scheme or use company staff
 - Cut down on expenses on consultants and professionals
 - Improve use of technology if possible
 - Benchmark current expenses with similar schemes and understand reasons for any excess

[Total Marks - 16]

Q.7)

Part (a)

- (i) A's decision will be driven primarily by which route(to sell/or not to sell) offers the best potential in terms of profit. In particular how the value it places on the business compares to the price B is prepared to pay. Or alternatively they could sell at a loss just to get rid of the hassle – ie intangible benefits.

For Company A

If the block is sold	If the block is not sold
There would be an immediate cash inflow, lump sum for the block sold which would include any value for future profits	- As it is closed for NB, the costs will have to be spread on a decreasing block of policies - Health insurance policies need

• hence may improve statutory solvency position • There could be potentially large cost savings as administering health reimbursement policies are more expensive than UL policies. For example: multiple claims, family may be covered under the same policy; claims investigation • After the line of business is sold it will leave a lot of staff redundant and such costs will need to be taken into account in the final transaction costs. • Selling could imply staff move as well. As it is a specialist area for A, the staff maybe needed by B then consideration of staff transfer costs • As it does not intend to continue this line of business, it can utilise all its resources towards its core business line. Might lead to more efficient business model. • Regulatory concerns on the transaction, impact on health policyholders in terms of service availability etc. • Likely Reaction from rating agencies and stock market if it is listed or proposing to get listed. • Likely use to which the cash inflow would be put to- new acquisitions, strengthening solvency, • Likely reaction of other policyholders- a sale could be perceived as financial weakness. • Time of sale (to get optimal value)	sophisticated administration. This system will need to be updated and/or upgraded, hence costs may continue to increase • Can these additional costs be passed on to the policyholders in the form of increased charges • could look at increasing premiums, however, this may cause lapses – which could be what company wants – or boost profits. • Whether the company will be able to increase the charges/premiums not only depends on the policy terms but also on the PRE and regulations. • Investment becomes messier and more difficult as fund size falls. • Look at alternative ways to boost profits eg cut expenses, tighten claims control. • May want to re-enter market later, which would be more difficult if sold. • Possible Undervaluation from stock market due to complex business.
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• Could be seen as having a less diversified portfolio compared to earlier position • If the sale is planned to raise cash and improve solvency position, are there any other more cost effective ways to raise money- financial reinsurance etc. • Could the cash received earn a better return than keeping the business would provide. • What about other purchasers – real or imagined to boost B's offer? • Any sale would have tax implications as well, which needs to be considered • Also would sale be a clean break ie any contingent liabilities from information not disclosed – sale contract needs careful drafting implying costs. • Would relations with brokers etc suffer impacting on other business?	
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For Company B

The main issue could be the profit it could make compared to what it has to pay to Company A.

- As the Company B is going to take over the risks in the health business; it is most important to be sure of the terms and conditions of the policies and charges Vs the risks taken on.
- Any options and guarantees will need to specially looked into and any costs associated with them will have to be explicitly factored
- Will need to identify any features that are significantly different from its existing business as this could potentially cause administrative difficulties and any such costs will have to be quantified
- There would be a onetime cost of migrating the Company A business into its system. This will need to be built into the transfer value.
- What are the gains for Company B from economies of scale, business synergy and wider geographical spread,

- vi. Since the transaction will be on cash basis, what is the optimal way to raise cash for the transaction? The cash transaction could impact free assets and its solvency position (especially if the business needs high reserves or significant one off costs post acquisition).
- vii. Could a stock cum cash transaction be more optimal,
- viii. what would be the regulatory concerns?
- ix. Consider likely reaction of all stakeholders-existing customers, new policyholders, rating agencies, shareholders and stock markets.
- x. Is this the best option acquisition option available currently eg any other possible targets.
- xi. especially look at timing issues and other options (not just this type of business) ie is this the right time to enter this market in terms of price v value
- xii. Could look at why A closed it – bad business with no prospects or A not really interested anymore.
- xiii. Is Company B looking at this purchase as a way to expand more ie to get a presence in areas where B is weak.
- xiv. Can current staff handle the transferred portfolio or is there requirement for hires(may be some from Company A itself) what then would be impact on existing staff morale.

Part(b)

Appropriate basis

- i. Starting point for both companies would be to project the cashflows-premiums, claims, expenses and investment proceeds, on best estimate basis for the closed block
- ii. Both companies will do this same exercise ie same method but different assumptions ie choice of assumptions will vary depending on different views of A and B
- iii. Best estimate basis for the closed block may be very different from current experience for Company B for similar product due to differences in target market, distribution channel used, geographical spread, underwriting standards. Moreover, this is a closed block of business with ageing policyholders and so may not have the experience expected eg selective lapses post acquisition of the closed block
- iv. Some assumptions may be correlated eg. premium changes vs selective lapses
- v. Even the recent past experience of company A might not be a good indicator of future experience as policyholders may react favorably/otherwise to the sale eg issues in iii above apply to A's calculations as well.
- vi. It is necessary to work out the value on a range of adverse scenarios
- vii. This will help them to understand the sensitivity of the assumptions and the potential financial impact of these factors

- viii. Thought there is potential expense savings for A and B, in most practical situations might not come true. Important to run the models on full expenses as well.
- ix. The claims experience might be very different from its own business (for Company B) due to differences in underwriting process and standards;
- x. Impact of adverse claims experience should be tested to check the sensitivity on the profits of the company.

[Total Marks – 18]

Q.8)

- a) Surplus or deficit is caused due to divergence in actual experience from valuation/reserving assumptions. The possible sources of surplus are:

- (i) Mortality
- (ii) Withdrawals/ lapses
- (iii) Investment return
- (iv) Expenses
- (v) Commission
- (vi) Expense Inflation
- (vii) Taxation
- (viii) Change in assumption compared to previous valuation.
- (ix) Options/guarantees biting,
- (x) Changes in statutory solvency reserve methodology ,
- (xi) Bonuses paid different to those assumed

b)

i. Mortality

Reasons for deterioration in experience compared to valuation assumptions

- a. Ineffective or poor policy and claims underwriting would lead to higher than expected mortality claims
- b. Fluctuations in mortality claims if portfolio size is small
- c. One off large claims -if the sum assured distribution is skewed and re- insurance is inadequate.
- d. Due to accumulations of risks and catastrophic events accompanied by inadequate reinsurance
- e. Unforeseen changes in mortality due to economic issues,
- f. New or more virulent diseases

ii. Withdrawals / lapses

Lapse/withdrawal experience could deteriorate due to-

- a. If the policyholders feel that their product is of lower value or
- b. Experience poor customer service levels
- c. New more attractive products in the market
- d. Attractive withdrawal benefits
- e. Change in tax regulations making insurance products unattractive
- f. General economic recession and policyholders need cash/have no money to pay premiums

iii. Investment return

Deterioration in investment experience could be due to

- a. Investment mismatch i.e. liabilities grow more than assets
- b. Poor investment management leading to relatively lower returns on free reserves or real liabilities
- c. General underperformance of markets
- d. Credit defaults
- e. Limited diversification across asset classes
- f. No hedge programs in place for protection against downside
- g. Wrong strategic benchmark used ie you thought that you were matched but you weren't

iv. Expenses

Deterioration in experience due to-

- a. Higher than expected expenses; due to inefficient management or
- b. underestimation of expense
- c. One off high expenses (like software purchase, voluntary retirement schemes) which cannot be amortized
- d. Fines or compensation payable
- e. Overuse of consultants
- f. New business strain or
- g. lower business volumes increasing unit costs

v. Commission

- a. Commission levels could go up from when priced; leading to increased expenses
- b. Change in distribution mix and product has been priced for a specific distribution mix/channel

- c. New distribution channel
- d. Changes in regulation leading to increase in commission payable,

vi. Inflation, expense and claim amounts

- a. Increase in inflation from expected would increase the actual expenses leading to higher unit costs than expected
- b. In particular this could arise from pay rises
- c. Due to say staff or skill shortages
- d. Or perhaps because expected cost savings don't come through
- e. Regulations such as health and safety or compliance could be an inflationary issue.

vii. Taxation

- a. The tax benefits available on policies are removed making such policies less attractive and so increasing lapse rates and higher unit costs than expected
- b. Decrease in new business volumes leading to higher unit costs than expected
- c. Tax charged on company profits could increase
- d. Particularly if overseas business is involved.

- viii. Change in assumption compared to previous valuation leading to one off surplus. Change in assumptions might be due to recent experience analysis, changes in regulations (expected to impact lapses) etc.
- ix. If the company offers financial options and guarantees- economic circumstances in the year might have been such making exercise of options profitable to the customer and hence increasing liability. This could impact surplus arising
- x. Changes in statutory solvency reserves methodology required- Recent changes in regulations making changes in valuation methodology necessary- for eg. Net premium to gross premium valuation method, changes in methodology for valuation of options and guarantees etc. This will impact surplus arising.
- xi. Bonuses paid different to those assumed- the actual bonus payouts could be different than assumed in valuation. For e.g., due to competitive pressure management may have decided to pay a higher bonus for claims occurring, one-off special payouts etc.

Part (c)

- i. Poor experience means less surplus – but high mortality on endowments could boost surplus if death benefits are relatively low, similarly the case with lapses
- ii. Less surplus means lower bonuses are possible and also less profit/dividend to any shareholders.
- iii. Depending on why the experience is poor- whether it is one off event or expected to persist company would have look at split between reversionary and terminal bonuses. A one off event can be reflected in lower TB while a persistent poor experience means regular bonuses have to be reduced.
- iv. Company may not want to cut bonuses but may try to smooth bonus payouts using free reserves or retained surplus set aside for this purpose or use any profits from without profit business to maintain bonus payouts.
- v. Any decision to maintain/reduce bonus payouts will depend on members' expectations, company's philosophy, past track record, sales illustrations and regulations.
- vi. Company may also be concerned about impact of lower bonuses on new business eg what are competition doing – are they in the same boat?
- vii. Company could also look at option of maintaining bonuses but increase premiums for new business but this implies cross-subsidy if written in the same pool.
- viii. Past record of company over last few years for e.g. have recent bonuses been relatively high so is lower surplus more normal?

[Total Marks – 20]

[Total Marks – 100]
